

Node.js



Model View Controller (MVC)

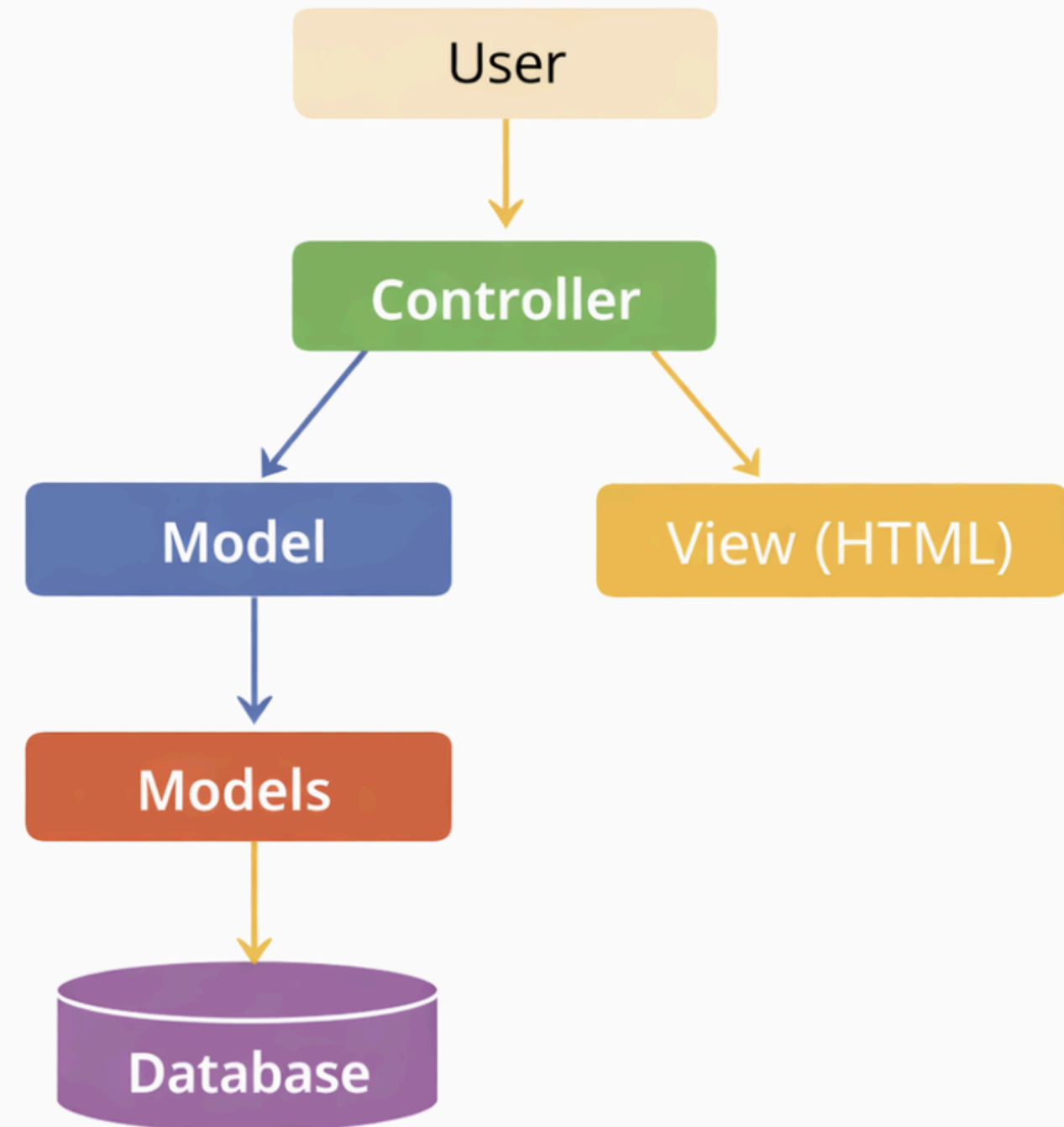
What is MVC?

- **MVC = Model - View - Controller**
- It is an architectural pattern that separates:
 - Model → Data & business rules
 - View → User interface
 - Controller → Handles user input

Classic MVC

- Controller handles request
- Model processes data
- View renders HTML page

CLASSIC MVC (TRADITIONAL WEB APP)



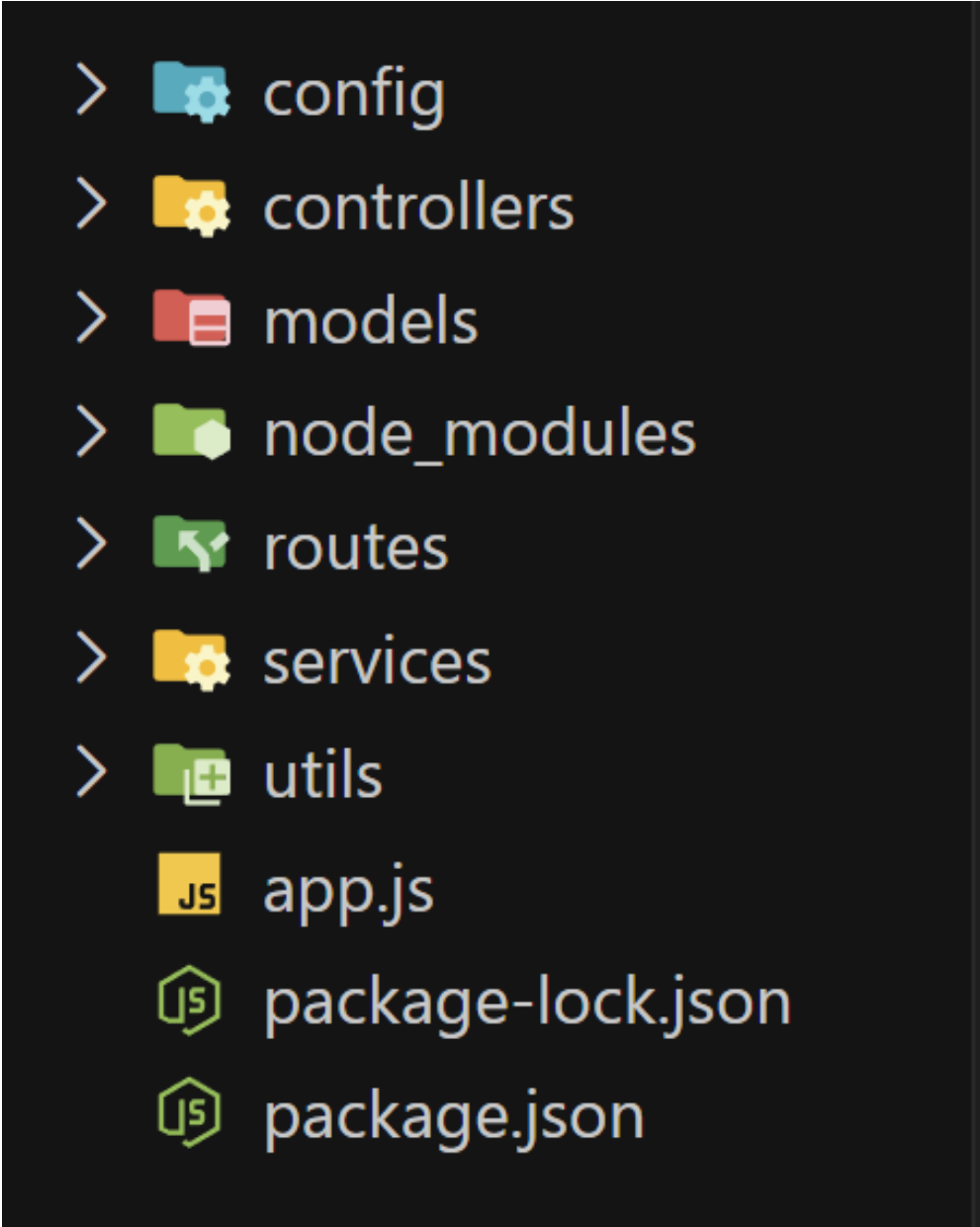
But What About RESTful API?

- **RESTful API** Does NOT Return HTML
- Instead of:
 - Returning View (HTML) we return **JSON Response**

```
{  
  "id": 1,  
  "name": "iPhone",  
  "price": 1200  
}
```

- 📌 JSON becomes the “View” in API architecture.

Project Structure



```
> config
> controllers
> models
> node_modules
> routes
> services
> utils
  app.js
  package-lock.json
  package.json
```

- Each folder has a responsibility.
- This is MVC + Service Layer.

Role of Each Layer

- **Routes**

- Define API endpoints
- Example: GET /products

- **Controllers**

- Handle req & res
- Call service
- Return JSON

- **Services**

- Business logic

Role of Each Layer

- Business logic
- Calculations
- Validation
- Rules
- **Models**
 - Database queries
 - SQL / ORM

Why Add Service Layer?

- Classic MVC puts business logic in Model.
- But modern apps:
 - Are larger
 - Have complex business rules
 - Need clean separation
 - May reuse logic

👉 Service Layer Pattern :

- Clean controllers
- Better scalability

Example

Model Layer Example

- Model:
 - Contains SQL
 - Talks directly to database

```
const db = require("../config/db");

async function getAll() {
  const [rows] = await db.query("SELECT * FROM products");
  return rows;
}

module.exports = { getAll };
```

Service Layer Example

- Service:
 - Business logic
 - No SQL
 - No HTTP logic

```
const productModel = require("../models/product");

async function getAllProducts() {
  return await productModel.getAll();
}

module.exports = { getAllProducts };
```

Controller Example

- Controller:
 - Handles request & response
 - Returns JSON

```
const productService = require("../services/product");

async function getProducts(req, res) {
  const products = await productService.getAllProducts();
  res.json(products);
}

module.exports = { getProducts };
```


Summary

- Modern REST API = **MVC + Service Layer Pattern**
- MVC is a concept, not a strict rule.
- In RESTful APIs:
 - **View** = JSON
 - **Service** = Optional but recommended
 - **Clean separation** = Professional backend design

